

Xiaoyu Zhang

CONTACT INFORMATION

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RESEARCH SUMMARY

My research focuses on the convergence of **wireless sensing**, **computational imaging**, and **smart health**. I turn low-cost commodity mmWave radar into a high-resolution imaging instrument for the human body, and the system built in my dissertation now images wounds *through their dressings* at a vascular surgery clinic under an IRB-approved protocol.

15 peer-reviewed papers (7 first-authored), **4 filed patents** (one under commercialization by a medical-device startup), and a **Best Paper Candidate Award**.

RESEARCH INTERESTS

I build sensing systems that reach real patients and real sites, along three lines:

- **mmWave sensing of the human body:** wound moisture and area imaging through dressings [IoTJ'25] [HumanSys'25], a deployed clinical station with longitudinal patient scanning, pixel-level hand localization from a single commodity radar [IoTJ'25], mmWave-voice identification [UIST'23], and a survey of mmWave for medicine [Sensors'25].
- **Multimodal medical AI:** chronic foot-ulcer prognosis from photoacoustic imaging [npj Imaging'26] [Photoacoustics'25], Alzheimer's screening from speech [ICIBM'26] [JBHI'24].
- **Sensing for sustainability:** robotic subsurface inspection of building thermal insulation [IROS'26].

EDUCATION

University at Buffalo, the State University of New York (SUNY) Sep. 2021 - Jul. 2026
Ph.D., Computer Science and Engineering Co-advised by Prof. Wenyao Xu & Yaxiong Xie
Dissertation (**defended Jul. 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
Committee: Wenyao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China Sep. 2017 - Jun. 2020
M.S., Electronics and Communication Engineering Advised by Prof. Bin Liu

Hefei University of Technology Sep. 2013 - Sep. 2017
B.Eng., Electronic Information Engineering

PUBLICATIONS

Accepted / published: 15 papers, 7 first-authored (name in **bold**). First-author work appears at IEEE IoTJ (CAS Tier-1 TOP), IEEE/RSJ IROS and ICIBM; co-authored work at ACM IMWUT, ACM UIST, npj Imaging, Photoacoustics and IEEE JBHI. Tags: [Wireless Sensing] [Medical AI] [Sustainability] .

[IROS'26] [Sustainability] **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wenyao Xu, "Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing", IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. (*Patent filed.*)

- [IMWUT’26] [Medical AI] Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wen Yao Xu, “*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*”, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.
- [ICIBM’26] [Medical AI] **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wen Yao Xu, “*Matching Speech Models to ADRD Subtypes: A Mechanistic Benchmark*”, International Conference on Intelligent Biology and Medicine, 2026.
- [npj Imaging’26] [Medical AI] Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wen Yao Xu, Jun Xia, “*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*”, npj Imaging, 2026.
- [HealthCom’26] [Medical AI] Nicholas Wei, Austin Xu, **Xiaoyu Zhang**, “*Pool or Specialize? Anatomic-Site Domain Shift in Melanoma Classification*”, IEEE International Conference on E-health Networking, Application & Services, 2026.
- [IoTJ’25] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wen Yao Xu, “*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*”, IEEE Internet of Things Journal, 2025. (Translated into US patent; under commercialization by a medical-device startup.)
- [IoTJ’25] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wen Yao Xu, “*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*”, IEEE Internet of Things Journal, 2025.
- [HumanSys’25] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wen Yao Xu, “*Through-dressing Wound Monitoring Based on the mmWave Sensor*”, The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (Best Paper Candidate)
- [Sensors’25] [Wireless Sensing] **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wen Yao Xu, “*A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications*”, Sensors, 2025.
- [Photoacoustics’25] [Medical AI] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wen Yao Xu, Jun Xia, “*Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging*”, Photoacoustics, 2025.
- [Photoacoustics’25] [Medical AI] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wen Yao Xu, Jun Xia, “*Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting*”, Photoacoustics, 2025.
- [JBHI’24] [Medical AI] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wen Yao Xu, “*A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer’s Disease Screening using Clock Drawing Test*”, IEEE Journal of Biomedical and Health Informatics, 2024.
- [UIST’23] [Wireless Sensing] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wen Yao Xu, Ming-Chun Huang, Kui Ren, “*WavoID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism*”, ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.
- [BSN’19] [Wireless Sensing] **Xiaoyu Zhang**, Bin Liu, “*A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs*”, IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.

[Bodynets'17] [Wireless Sensing] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, “JMMM: A Mobility Model for WBANs Based on Human Joint Movements”, 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

PATENTS

4 patents filed (US). Each corresponds to a first-authored system paper or a deployed prototype.

- **Device and Method for Wound Monitoring and Diagnosis using Radio-Frequency Technologies**. *Corresponds to mmSkin (IoTJ'25)*. [Under commercialization by a medical-device startup](#).
- **Autonomous Robotic System for Non-Contact Volumetric Thermal Conductivity Mapping Using Millimeter-Wave Synthetic Aperture Radar**. *Corresponds to IROS'26*.
- **Vision-Based Closed-Loop Control System for Autonomous Biomass Drying with Predictive Crack Prevention**
- **A New mmWave System for Accurate Hand and Finger Tracking**. *Corresponds to mm-Hand (IoTJ'25)*.

HONORS AND AWARDS

- **Best Paper Candidate Award**, HumanSys'25. The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair's Fellowship**, Department of CSE, University at Buffalo, 2021
- Merit scholarships (graduate and undergraduate), USTC and HFUT, 2015–2019

INDUSTRY & CLINICAL EXPERIENCE

Clinical Research Collaborations 2022 - Present
Multi-year IRB-approved clinical collaborations at the University at Buffalo and partner sites, spanning device deployment, on-site data collection, and multi-modal cohort curation:

- **Wound care** (UBMD Vascular Surgery). Designed, built and deployed a mmWave clinical wound-imaging station on site; **5 patients** scanned to date under an IRB-approved protocol, with repeat longitudinal scanning up to **3 months** per patient
- **Breast imaging**. Clinical cohort of **284 imaged patients**; curated the **53-patient** tri-modal photoacoustic / ultrasound / mammography subset used for molecular subtyping
- **Wearable ear-EEG**. **27 subjects** recorded across four passive auditory tasks (~1,800 trials), **23** retained after per-channel quality control, paired with standardized clinical score scales
- **Thyroid ultrasound**. Ongoing clinical scanning collaboration for nodule diagnosis and lesion segmentation
- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering). Joint imaging-system development and radiomics analysis

China Merchants Bank Software Center, Hangzhou, Zhejiang, China Jul. 2020 - Sep. 2020
Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

COMPETITION

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** Mar. 2025
Advanced Through-Gauze Wound Assessment Technology.
Xiaoyu Zhang (technical lead and presenter), Chuhui Liu, Jun Xia, Linda M. Harris, Wen Yao Xu.
[Advanced to the semifinals](#); pitched the patented mmWave wound-assessment platform to clinical and industry judges.

PRESENTATIONS
& DEMOS

- **Ph.D. Dissertation Defense:** *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications* Jul. 2026
University at Buffalo, CSE
- **Live Demonstration:** SUNY Research Expo Mar. 2026
mmWave clinical wound-imaging station, translational outreach
- **Community Engagement Studio:** UB CTSI Feb. 2026
Wound-measurement device design review with a 12-member community expert panel
- **Invited Talk:** The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk:** Panasci Technology Entrepreneurship Competition Mar. 2025
Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk:** IEEE-EMBS International Conference on Body Sensor Networks May 2019
A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk:** 19th EAI International Conference on Body Area Networks Sep. 2017
JMMM: A Mobility Model for WBANs Based on Human Joint Movements

TEACHING
EXPERIENCES

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024
2. Special Topics, UB. **Invited guest lecture:** *Wireless Signal Processing: Making Sense of the Invisible* Fall 2023
3. Mathematical Logic and Graph Theory, USTC Fall 2018

MENTORING
EXPERIENCES

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects. **Four high-school students I mentored are first authors on peer-reviewed papers.**

- Prakriti Setlur, AI-based clinical image screening pipelines
- Kh Shafin Farhan, foundation models for medical imaging
- Abhi Ramtel (CSE@UB), UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB), 3D model design of the mmWave sensor scanning system
- George Gillman (EE@UB), mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB), embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB), tooling for automatic image labeling
- Nicholas Wei & Austin Xu (high school), anatomic-site domain shift in melanoma classification
- Henry Yang & Emily Liu (high school), real-world study of sport-based support for autism spectrum disorder

COMMUNITY
SERVICES &
OUTREACH
ACTIVITIES

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2024, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo**. Translational outreach for the mmWave clinical wound-imaging station [Mar. 2026]
- **Community Engagement Studio**, UB Clinical and Translational Science Institute (CTSI). Convened a **12-member community expert panel** (healthcare providers, clinical trainees, community advocates) for a design review of the wound-measurement device [Feb. 2026]